Predictive Maintenance and Unplanned Downtime

Abstract. We study the effect of predictive maintenance on unplanned downtime in discrete manufacturing plants. Across twelve plants, downtime fell by thirty one percent.

1. Introduction

Unplanned downtime remains the largest single source of lost output in discrete manufacturing. Condition monitoring is widely deployed but rarely linked to maintenance planning.

2. Method

We instrumented 214 machines with vibration and temperature sensors and trained a gradient boosted model to predict bearing failure seven days ahead.

3. Results

Table 1 reports downtime before and after the programme.

Plant group	Machines	Downtime before (h)	Downtime after (h)
Pumps	82	1,420	960
Gearboxes	64	1,115	790
Compressors	68	1,360	905

Table 1. Unplanned downtime by plant group.

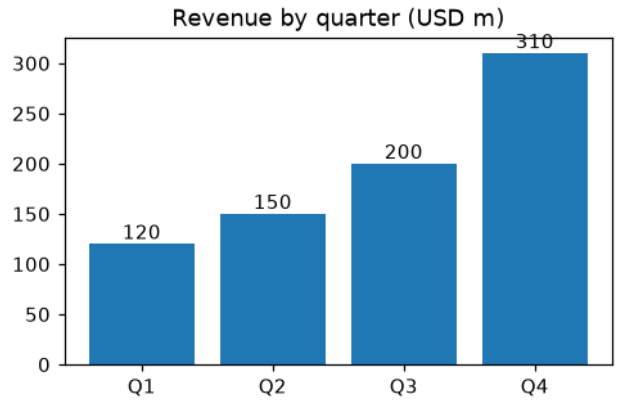


Figure 1. Revenue by quarter (USD m).

4. Conclusion

Predictive maintenance reduced downtime in every plant group.